Caleb James Smith

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Summary

Software developer and data scientist with a background in experimental high-energy physics. Ten years of experience in physics research as a collaborator on the Compact Muon Solenoid (CMS) experiment. Experienced, persistent, and creative problem solver. Hardworking team player. Curious learner, researcher, and teacher.

Technical Strengths

Languages	Python, C++, Bash, HTML, CSS
Libraries	Matplotlib, NumPy, Pandas, Flask, Django, ROOT
Tools	Unix/Linux Terminal, VS Code, Vim, Git, JSON, XML, Markdown, LaTeX

Work Experience

AI Evaluator (Part Time) — Feb. 2026–Present
DataAnnotation — *Remote*

- Reviewed and graded the quality of AI responses (text output from LLMs). Created custom, realistic system and user prompts and evaluated model responses.

Postdoctoral Researcher (Physics) — Jan. 2021–Jan. 2026
The University of Kansas — *Lawrence, KS*

- Analyzed CMS data and Monte Carlo simulation (billions of events) using over 2,000 independent search regions in a data-driven fit to search for supersymmetric particles. Used ML algorithms (DNNs) to identify bottom quarks.
- Developed an automated electrical testing framework (Python) to test high-speed (1.28 Gbps) electrical cables (e-links). Reduced testing time by a factor of 10. Framework used to test over 1,000 e-links for the CMS pixel tracker upgrade.
- Created a visualization of e-link production for each stage over time (Python, Pandas, NumPy, Matplotlib). Demonstrated that e-link production was projected to finish 1.5 years ahead of schedule.

Research Assistant (Physics) — May 2016–Dec. 2020
Baylor University — *Fermilab (Batavia, IL) and CERN (Meyrin, Switzerland)*

- Analyzed CMS data and Monte Carlo simulation (billions of events) to search for supersymmetric particles. Used data-driven analysis to predict the Z to neutrinos background and associated uncertainties. Used ML algorithms (DNNs) to identify top/bottom quarks and W bosons.
- Created and maintained software with a team at Fermilab, including GUI, database, and website (Python, Tkinter, Django, Bash), to test over 700 electronic readout cards for the CMS hadron calorimeter upgrade.

Teaching Assistant (Physics), Instructor of Record — Aug. 2014–May 2016
Baylor University — *Waco, TX*

- Taught undergraduate lectures (Physics 2) and labs (Physics 1 and 2).

Personal Projects

Chess program	Interactive chess program for human and computer players built in Python using PyGame.
Speedcubing app	Rubik's Cube algorithm Android app built using Android Studio, Kotlin, and Jetpack Compose.

Education

Baylor University — Aug. 2014–Dec. 2020
PhD in Physics
Dissertation: *Search for supersymmetric top quarks in the CMS Run 2 data set*

Baylor University — Aug. 2014–May 2018
MA in Physics

Taylor University — Aug. 2011–May 2014
BS in Physics, Minor in Mathematics